

R workshop part 5 - Paths, projects, and data import

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Projects & data import presentation



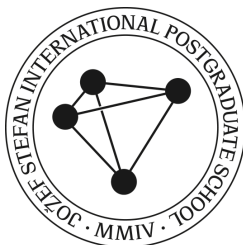
Paths

Paths are directions that tell R where to look for data or place outputs on our computer.

- Example Windows path: `C:\Users\gacnik\Desktop`
- Example Mac path: `/Users/gacnik/Desktop`

R only accepts paths with forward slash character (`/`), meaning that:

- Windows paths cannot be used as is, replacement with `/` characters
- Mac paths can be used as is



Paths

Use `getwd()` to **get** your *current* R working directory. This is where R currently looks for data and saves outputs to.

```
1 getwd()
```

```
[1] "C:/Users/gacnik/OneDrive - ijs.si/Jan 2023+/R course/Data_analysis_in_R"
```

Use `setwd()` to **set** your *new* R working directory.

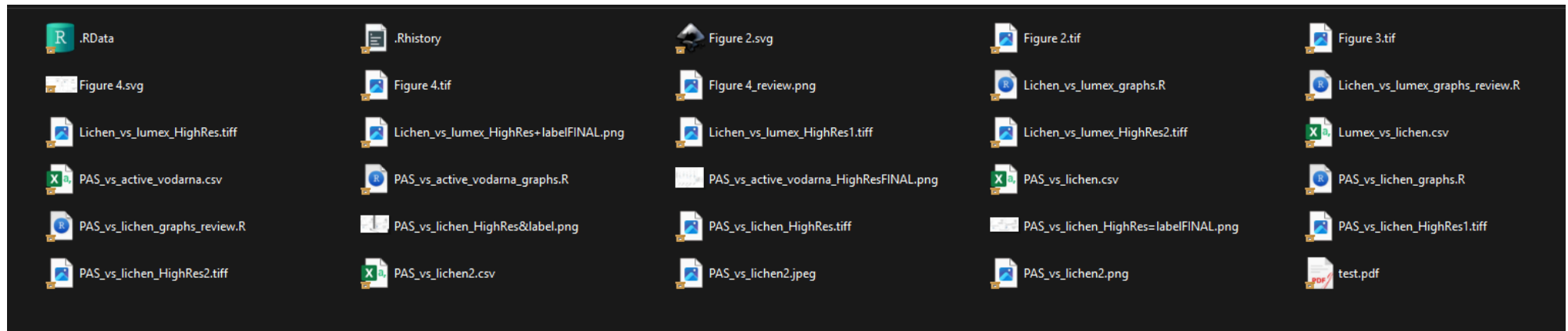
```
1 setwd("C:/ENTER/YOUR/PATH/HERE")
```

There is an easier and better way to manage paths through **RStudio projects**

Organizing your analysis project

As your data analysis skill improve, tasks become larger, and supporting files start growing.

- We started our work in an R script, then we generated some plots, tables... if we started saving these outputs and adding new data, things could get disorganized fast:



Personal archive, my first messy R project

Organizing your analysis project

Project organization for larger tasks/projects:

- everything in **one project folder**, with standard **sub folders**
 - subfolders `data`, `scripts`, `outputs`, ...
- use of **RStudio project file** (`.Rproj` file)
 - relative instead of absolute paths
 - preserves specific rstudio environment and setup just for the project

(Showcase the use of RStudio project)

Relative paths make projects portable

We said RStudio project uses relative paths, not absolute, what does it mean?

- **Absolute path:** starts from a fixed location on one computer

```
C:/Users/gacnik/Desktop/Example_project/data/penguins.csv
```

- **Relative path:** starts from the project folder

```
data/penguins.csv
```

Rule: Always use relative paths in project scripts. Avoid hard-coded absolute paths: they usually break when a project is moved, shared, or opened on another computer.

Data import

Working with data already provided by R was neat, but you will need to apply the learned concept on your data.

Here, we will focus on plain-text rectangular files such as CSV, TSV, and Excel files

```
Code,Station name,Units,Observed conc,Modelled conc
DE0002R,Waldhof,ng/m3,1.58,1.23
DE0003R,Schauinsland,ng/m3,1.24,1.21
DE0009R,Zingst,ng/m3,1.45,1.29
EE0009R,Lahemaa,ng/m3,1.10,1.25
FI0027R,Virolahti Harju,ng/m3,1.16,1.22
FI0050R,Hyytiala,ng/m3,1.24,1.18
GB0048R,Auchencorth Moss,ng/m3,1.29,1.22
GB1055R,Chilbolton Observatory,ng/m3,1.42,1.28
NO0042G,Zeppelin mountain ,ng/m3,1.53,1.13
NO0073U,Sofienbergparken,ng/m3,1.50,1.18
PL0005R,Diabla Gora,ng/m3,1.37,1.19
SE0014R,Rao,ng/m3,1.26,1.26
```

Example of a simple CSV file

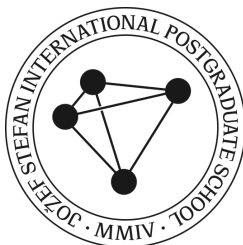
Data import

Working with data already provided by R was neat, but you will need to apply the learned concept on your data.

Here, we will focus on plain-text rectangular files such as CSV, TSV, and Excel files

Code	Station name	Units	Observed conc	Modelled conc
DE0002R	Waldhof	ng/m3	1.58	1.23
DE0003R	Schauinsland	ng/m3	1.24	1.21
DE0009R	Zingst	ng/m3	1.45	1.29
EE0009R	Lahemaa	ng/m3	1.1	1.25
FI0027R	Virolahti Harju	ng/m3	1.16	1.22
FI0050R	Hyytiala	ng/m3	1.24	1.18
GB0048R	Auchencorth Moss	ng/m3	1.29	1.22
GB1055R	Chilbolton Observatory	ng/m3	1.42	1.28
NO0042G	Zeppelin mountain	ng/m3	1.53	1.13
NO0073U	Sofienbergparken	ng/m3	1.5	1.18
PL0005R	Diabla Gora	ng/m3	1.37	1.19
SE0014R	Rao	ng/m3	1.26	1.26

Same example, viewed as a table



Data import with a relative path

1. Download and extract the prepared [Example RStudio project](#).
2. Open `Example_project.Rproj`. Then import the file with `read_csv()` using a relative path:

```
1 library(tidyverse)
2
3 mercury_air <- read_csv("data/mercury_air.csv")
```

```
# A tibble: 12 × 5
  Code      `Station name`  Units `Observed conc` `Modelled conc`
  <chr>    <chr>          <chr> <chr>              <dbl>
1 DE0002R Waldhof        ng/m3 1.58            1.23
2 DE0003R N/A            ng/m3 1.24            1.21
3 DE0009R Zingst         ng/m3 1.45            1.29
4 EE0009R Lahemaa        ng/m3 N/A          1.25
5 FI0027R Virolahti Harju ng/m3 1.16            1.22
6 FI0050R N/A            ng/m3 1.24            1.18
7 GB0048R Auchencorth Moss ng/m3 1.29            1.22
# i 5 more rows
```



Data import

`read_csv()` function read the N/A cells as a text, not as a true NA value. We need to define what represents NA values using the `na` argument:

```
1 mercury_air <- read_csv("data/mercury_air.csv", na = c("N/A", ""))
2 mercury_air
```

A tibble: 12 × 5

	Code	Station name	Units	Observed conc	Modelled conc
	<chr>	<chr>	<chr>	<dbl>	<dbl>
1	DE0002R	Waldhof	ng/m3	1.58	1.23
2	DE0003R	<NA>	ng/m3	1.24	1.21
3	DE0009R	Zingst	ng/m3	1.45	1.29
4	EE0009R	Lahemaa	ng/m3	NA	1.25
5	FI0027R	Virolahti Harju	ng/m3	1.16	1.22
6	FI0050R	<NA>	ng/m3	1.24	1.18
7	GB0048R	Auchencorth Moss	ng/m3	1.29	1.22
8	GB1055R	Chilbolton Observatory	ng/m3	1.42	1.28



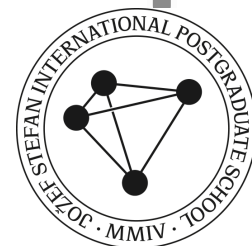
Data import

The names of columns have white space, and as it is annoying to manually rename each one using the `rename()` function. We can do it using the janitor package function `clean_names()`:

```
1 mercury_air <- read_csv("data/mercury_air.csv", na = c("N/A", "")) %>%
2   clean_names()
3
4 mercury_air
```

A tibble: 12 × 5

	code	station_name	units	observed_conc	modelled_conc
	<chr>	<chr>	<chr>	<dbl>	<dbl>
1	DE0002R	Waldhof	ng/m3	1.58	1.23
2	DE0003R	<NA>	ng/m3	1.24	1.21
3	DE0009R	Zingst	ng/m3	1.45	1.29
4	EE0009R	Lahemaa	ng/m3	NA	1.25
5	FI0027R	Violahti Harju	ng/m3	1.16	1.22
6	FI0050R	<NA>	ng/m3	1.24	1.18
7	GB0048R	Auchencorth Moss	ng/m3	1.29	1.22
8	GB1055R	Chilbolton Observatory	ng/m3	1.42	1.28



Data import

Depending on the mess level of your data files, more arguments of `read_csv()` could be needed, for example:

- using the `skip` argument to tell `read_csv()` how many metadata headers in files to **skip**
- using the `col_types` argument to explicitly tell `read_csv()` what type each column should be, instead of relying on guessing
- using one of `parse_number()`, `parse_date()`, `parse_time()`, ... functions to **convert values in columns to their correct type**



Import several similar files at once

The `data/` folder also contains similarly structured files per year:

`mercury_air_2022.csv`, `mercury_air_2023.csv`,
`mercury_air_2024.csv`

- Let's make a vector - `c()` - of file names, and pass it to the `read_csv()`:

```
1 mercury_files <- c("data/mercury_air_2022.csv",  
2                   "data/mercury_air_2023.csv",  
3                   "data/mercury_air_2024.csv")  
4  
5 mercury_air_all <- read_csv(mercury_files, id = "file_name")  
6 mercury_air_all
```

Import several similar files at once

The result is a data frame which contains data from all three CSV files:

```
# A tibble: 36 × 7
```

	file_name	Year	Code	`Station name`	Units	`Observed conc`	`Modelled conc`
	<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>
1	data/mercur...	2022	DE00...	Waldhof	ng/m3	1.55	1.2
2	data/mercur...	2022	DE00...	N/A	ng/m3	1.18	1.17
3	data/mercur...	2022	DE00...	Zingst	ng/m3	1.4	1.26
4	data/mercur...	2022	EE00...	Lahemaa	ng/m3	1.08	1.21
5	data/mercur...	2022	FI00...	Violahti Har...	ng/m3	1.12	1.18
6	data/mercur...	2022	FI00...	N/A	ng/m3	1.2	1.15
7	data/mercur...	2022	GB00...	Auchencorth M...	ng/m3	1.25	1.18
8	data/mercur...	2022	GB10...	Chilbolton Ob...	ng/m3	1.38	1.24
9	data/mercur...	2022	N000...	Zeppelin moun...	ng/m3	1.49	1.1
10	data/mercur...	2022	N000...	Sofienbergpar...	ng/m3	1.46	1.16
11	data/mercur...	2022	PL00...	Diabla Gora	ng/m3	N/A	1.16

Data export

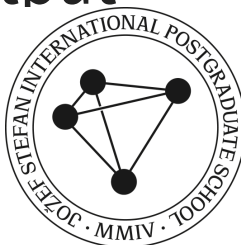
Data export or data writing is the reverse process of data import. Data can be saved to CSV or TSV using `write_csv()` and `write_tsv()`, respectively.

- Two arguments are used: `x` - data frame to be saved, and `file` - the location where to save it:

```
1 write_csv(mercury_air_all, "outputs/mercury_air_all.csv")
```

Note that the file was saved into the `outputs` sub directory of our project directory

- Handy for saving text file outputs, figures, models, or any other output results.



Data entry

Sometimes we need to make a table “by hand” using `data.frame()`. This can be done by data entry in your R script, although it is not recommended for larger data.

- The layout follows this structure: `column_name1 = c(value1, value2, value3)`, `column_name2 = c(value4, value5, value6)`, ... = ...

```
1 data.frame(  
2   x = c(1, 2, 4),  
3   y = c("dog", "cat", "pig"),  
4   z = c(0.02, 0.05, 0.08)  
5 )
```

	x	y	z
1	1	dog	0.02
2	2	cat	0.05
3	4	pig	0.08

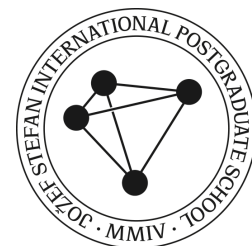
Part 1 · Introduction (1/2)

Work in a script

- Run the current line with `Ctrl` + `Enter`
- Use `#` comments to explain **why**
- Use `?function_name` when you need help
- Read errors from the first line and check brackets, commas, and spelling

Think in functions and objects

- Functions perform actions: `mean(x)`
- Arguments control them: `mean(x, na.rm = TRUE)`
- `<-` stores a result in an object
- Clear names make later steps easier to follow



Part 1 · Introduction (2/2)

```
1 # Install once; load each session
2 install.packages("tidyverse")
3 library(tidyverse)
4
5 mass <- c(4200, 3900, 4600)
6 mean_mass <- mean(mass)
```

- A vector contains values of one basic type
- A data frame organizes vectors into columns
- Use `object[row, column]` to subset a data frame
- Installing adds a package; loading makes its functions available
- Reassigning an existing name overwrites the object

Part 2 · Visualization (1/2)

Data → aesthetic mappings → geometric layers

```
1 ggplot(  
2   data = penguins,  
3   mapping = aes(  
4     x = body_mass,  
5     y = flipper_len,  
6     color = species  
7   )  
8 ) +  
9   geom_point()
```

- Put variables inside `aes()`
- Put fixed choices outside `aes()`
- Add layers with `+`
- Choose a geometry that answers the question
- Add labels and a theme only after the plot communicates clearly

Part 2 · Visualization (2/2)

What do you want to see?

- One numeric variable → histogram or density plot
- Numeric values across groups → boxplot
- Two numeric variables → scatter plot
- Patterns within groups → color, shape, or facets

```
1 penguins %>%  
2   ggplot(aes(  
3     x = body_mass,  
4     y = flipper_len,  
5     color = species  
6   )) +  
7   geom_point() +  
8   facet_wrap(~ sex) +  
9   labs(  
10    x = "Body mass (g)",  
11    y = "Flipper length (mm)"  
12  ) +  
13  theme_minimal()
```

Save the final plot with `ggsave()`; set its filename, size, and resolution.

Part 3 · Data transformation (1/2)

Goal	Main verbs
Keep rows	<code>filter()</code> , <code>slice()</code> , <code>distinct()</code>
Work with columns	<code>select()</code> , <code>rename()</code> , <code>mutate()</code>
Count observations	<code>count()</code>
Calculate by group	<code>group_by()</code> + <code>summarise()</code>

Before summarising: check missing values; use `na.rm = TRUE` when removal is appropriate.

Part 3 · Data transformation (2/2)

```
1 penguins %>%
2   filter(!is.na(sex), !is.na(body_mass)) %>%
3   mutate(body_mass_kg = body_mass / 1000) %>%
4   group_by(species, sex) %>%
5   summarise(
6     mean_mass_kg = mean(body_mass_kg),
7     n = n(),
8     .groups = "drop"
9   )
```

Read top to bottom: **start with data**, then add one clear step per line.

Part 4 · Data tidying (1/2)

- Each **variable** is a column
- Each **observation** is a row
- Each **value** is one cell

Tidy data makes it easier to:

- transform with `dplyr`
- visualize with `ggplot2`
- compare observations
- reuse code across variables and groups



Part 4 · Data tidying (2/2)

Wide → long

```
1 billboard %>%  
2   pivot_longer(  
3     cols = starts_with("wk"),  
4     names_to = "week",  
5     values_to = "rank",  
6     values_drop_na = TRUE  
7   )
```

Long → wide

```
1 Orange %>%  
2   pivot_wider(  
3     names_from = age,  
4     values_from = circumference,  
5     names_prefix = "age_days_"  
6   )
```

Use consistent spacing and indentation; break long pipelines and named arguments across lines.

Part 5 · Paths, projects, and data import

```
1 my_project/  
2 | my_project.Rproj  
3 | data/  
4 | scripts/  
5 | outputs/
```

- Open the `.Rproj` file before working
- Keep inputs, scripts, and outputs in predictable folders
- Use relative paths such as `data/file.csv`
- Avoid paths tied to one person or computer
- Treat original data as read-only

Test: move the project folder; the scripts should still run.

